DESCENT DATA FOR AFFINE MORPHISMS OVER AN FPQC COVERING ARE EFFECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Effectivity of descent data for morphisms whose type descends, over an fpqc covering. Same chapter, Lemma lemma-descending-types-morphisms, invoked at proof lines 9380-9383. Reduces the general fpqc covering to a single flat surjective morphism of affines.
- Characterisation of affine morphisms and base-change stability. Morphisms, Lemmas morphisms-lemma-characterize-affine and morphisms-lemma-base-change-affine, invoked at proof lines 9376-9379. Allows the reduction and writes V = Spec(B) via Morphisms, Definition morphisms-definition-affine.
- Effectivity of descent for modules along a faithfully flat ring map. Same chapter, Proposition proposition-descent-module, invoked at proof lines 9406-9412. Produces the R-module C with A tensor_R C = B compatible with the descent data.

Source reference. Original lemma 02W3: sources/stacks-descent/pinned-original-source.tex, source lines 9245–9245. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 01S6: sources/stacks-chow/tags.txt, source lines 2327–2327. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01SD: sources/stacks-chow/tags.txt, source lines 2334–2334. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 01S8: sources/stacks-chow/tags.txt, source lines 2329–2329. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original proposition 023N: sources/stacks-descent/pinned-original-source.tex, source lines 859–859. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 022B: sources/stacks-chow/tags.txt, source lines 2700–2700. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let $R \to A$ be a ring map.

(1) A *descent datum* (N, φ) *for modules with respect to* $R \to A$ is given by an A-module N and an isomorphism of $A \otimes_R A$-modules

$$\varphi : N \otimes_R A \to A \otimes_R N$$

such that the *cocycle condition* holds: the diagram of $A \otimes_R A \otimes_R A$-module maps

$$\begin{array}{ccc} N \otimes_R A \otimes_R A & \xrightarrow{\varphi_{02}} & A \otimes_R A \otimes_R N \\ & {\scriptstyle \varphi_{01}} \searrow \quad \nearrow {\scriptstyle \varphi_{12}} & \\ & A \otimes_R N \otimes_R A & \end{array}$$

commutes (see below for notation).

(2) A *morphism* $(N, \varphi) \to (N', \varphi')$ *of descent data* is a morphism of A-modules $\psi : N \to N'$ such that the diagram

$$\begin{array}{ccc} N \otimes_R A & \xrightarrow{\varphi} & A \otimes_R N \\ {\scriptstyle \psi \otimes \mathrm{id}_A} \downarrow & & \downarrow {\scriptstyle \mathrm{id}_A \otimes \psi} \\ N' \otimes_R A & \xrightarrow{\varphi'} & A \otimes_R N' \end{array}$$

is commutative.

Lemma 0.2. *Let S be a scheme. Let $\{X_i \to S\}_{i \in I}$ be an fpqc covering, see Topologies, Definition 022B. Let $(V_i / X_i, \varphi_{ij})$ be a descent datum relative to $\{X_i \to S\}$. If each morphism $V_i \to X_i$ is affine, then the descent datum is effective.*

Proof. Being affine is a property of morphisms of schemes which is local on the base and preserved under any base change, see Morphisms, Lemmas 01S8 and 01SD. Hence Lemma 02W3 applies and it suffices to prove the statement of the lemma in case the fpqc-covering is given by a single $\{X \to S\}$ flat surjective morphism of affines. Say $X = \mathrm{Spec}(A)$ and $S = \mathrm{Spec}(R)$ so that $R \to A$ is a faithfully flat ring map. Let (V, φ) be a descent datum relative to X over S and assume that $V \to X$ is affine. Then $V \to X$ being affine implies that $V = \mathrm{Spec}(B)$ for some A-algebra B (see Morphisms, Definition 01S6). The isomorphism φ corresponds to an isomorphism of rings

$$\varphi^\sharp : B \otimes_R A \longleftarrow A \otimes_R B$$

as $A \otimes_R A$-algebras. The cocycle condition on φ says that

$$\begin{array}{ccc} B \otimes_R A \otimes_R A & \longleftarrow & A \otimes_R A \otimes_R B \\ & \nwarrow \quad \swarrow & \\ & A \otimes_R B \otimes_R A & \end{array}$$

is commutative. Inverting these arrows we see that we have a descent datum for modules with respect to $R \to A$ as in Definition 0.1. Hence we may apply Proposition 023N to obtain an R-module $C = \mathrm{Ker}(B \to A \otimes_R B)$ and an isomorphism $A \otimes_R C \cong B$ respecting descent data. Given any pair $c, c' \in C$ the product cc' in B lies in C since the map φ is an algebra homomorphism. Hence C is an R-algebra whose base change to A is isomorphic to B compatibly with descent data. Applying Spec we obtain a scheme U over S such that $(V, \varphi) \cong (X \times_S U, can)$ as desired. $\square$